

Ohm's Law: The Rule That Runs Every Circuit

Basic Electrical and Electronics Engineering - Unit 1: Electric Circuits - Session 2

1. Learning objectives

- Define voltage, current and resistance and state their SI units.
- State Ohm's law and apply $V = IR$ to find an unknown quantity.
- Interpret a V-I graph to determine resistance.
- Distinguish ohmic from non-ohmic devices.

2. Voltage, current and resistance

Electric current (I) is the rate of flow of electric charge through a conductor, $I = Q / t$. Its SI unit is the ampere (A): one ampere is one coulomb of charge passing a point every second.

Voltage (V), or potential difference, is the work done per unit charge to move charge between two points, $V = W / Q$. Its unit is the volt (V), equal to one joule per coulomb.

Resistance (R) is the opposition a material offers to the flow of current, measured in ohms (Ω). It depends on the material's resistivity ρ , the length L and the cross-sectional area A of the conductor: $R = \rho L / A$. A longer wire resists more; a thicker wire resists less.

Quantity	Symbol	SI unit	Unit symbol
Current	I	ampere	A
Voltage	V	volt	V
Resistance	R	ohm	Ω
Power	P	watt	W

3. Ohm's law

At constant temperature, the current through a conductor is directly proportional to the potential difference across its ends. The constant of proportionality is the resistance.

$$V = I \times R$$

Rearranged: $I = V / R$ and $R = V / I$. Doubling the voltage across a fixed resistor doubles the current; doubling the resistance at a fixed voltage halves the current.

4. Experimental verification

A $20\ \Omega$ resistor was connected to a variable DC supply with an ammeter in series and a voltmeter in parallel. The readings were:

Voltage V (V)	Current I (A)	V / I (Ω)
2.0	0.10	20
4.0	0.20	20
6.0	0.30	20
8.0	0.40	20
10.0	0.50	20

The V-I graph is a straight line through the origin; its slope $V / I = 20\ \Omega$ is the resistance.

5. Worked example

A 12 V battery is connected across a $4\ \Omega$ resistor. Find the current and the power dissipated.

1. Given: $V = 12\text{ V}$, $R = 4\ \Omega$.
2. Ohm's law: $I = V / R = 12 / 4 = 3\text{ A}$.
3. Power: $P = V \times I = 12 \times 3 = 36\text{ W}$ (check: $P = I^2R = 9 \times 4 = 36\text{ W}$).

6. A common misconception

Many students believe a resistor 'uses up' current, so less current leaves it than enters. In a series loop the current is the same everywhere. The resistor converts electrical energy into heat; that energy transfer appears as a drop in voltage across the resistor, not as a loss of current.

7. Limitations

Ohm's law holds for metallic conductors at constant temperature (ohmic devices). Filament lamps (resistance rises as the filament heats up), diodes and thermistors are non-ohmic: their V-I graphs are curved, so $R = V / I$ changes with the operating point.

8. Practice

1. A 230 V room heater draws 5 A. What is its resistance?
2. What current flows through a $1\text{ k}\Omega$ resistor connected to a 9 V battery?
3. The current through a lamp doubles when the voltage is tripled. Is the lamp ohmic? Explain.